

# Solution - Bitmaps

Periodical compact/defragment disk

- locks up disk bandwidth during operation

Keep adjacent free blocks together on freelist

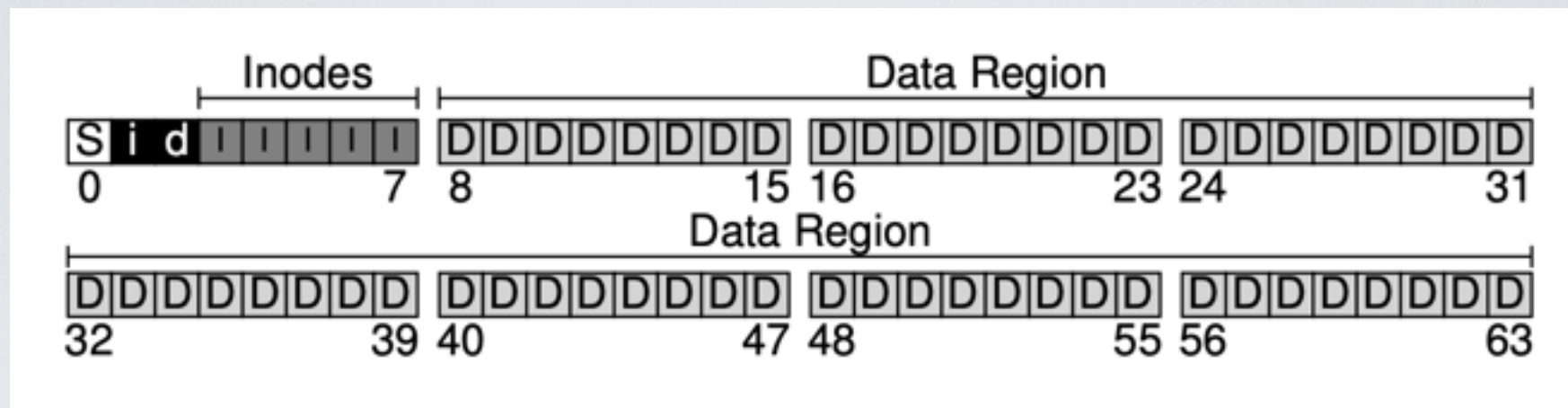
- costly to maintain

➔ FFS - bitmap of free blocks (same idea as Indexed File System)

- Each bit indicates whether block is free  
e.g., 10101011111100000111111000101100
- Easier to find contiguous blocks
- Small, so usually keep entire thing in memory
- Time to find free blocks increases if fewer free blocks

|                  |      |      |      |       |
|------------------|------|------|------|-------|
| Bits in map      | XXXX | XXOO | OOXX | OOOO  |
| Fragment numbers | 0-3  | 4-7  | 8-11 | 12-15 |
| Block numbers    | 0    | 1    | 2    | 3     |

# Unix File System



The disk is (physically) divided into sectors (usually 512 bytes per sector)

The file system is (logically) divided into blocks (e.g., 4 KB per block)

➔ Disk space is allocated in granularity of blocks

1. The data blocks "D" stored files (and directories) content
2. The inodes blocks "I" stores the inode table
3. The data bitmap "d" block d tracks which data block is free or allocated (one bit per block on the disk)
4. The inode bitmap "i" block i tracks which inode is free or allocated (one bit per inode)
5. The Superblock "S" (a.k.a Master Block or partition control block) contains:
  - a magic number to identify the file system type
  - the number of blocks dedicated to the two bitmaps and inodes